Wealth Vault Scoring Engine Specification

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1 Inputs

For each stock i, as of date T:

1.1 Price data

- Daily adjusted close prices $P_{i,t}$ for at least the last 252 trading days (1 year).
- Daily traded value (for liquidity checks, if needed later).

Derived.

Daily simple return

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}. \tag{1}$$

Cumulative returns

For a window of length W:

$$\mathrm{Ret}_{i,W} = \prod_{k=1}^{W} \big(1 + R_{i,t-k+1}\big) - 1. \tag{2}$$

Required windows

$$W_{1M} = 21 \text{ trading days}, \qquad \mathrm{Ret}_{i,1M} = \mathrm{Ret}_{i,W_{1M}}, \tag{3}$$
$$W_{3M} = 63 \text{ trading days}, \qquad \mathrm{Ret}_{i,3M} = \mathrm{Ret}_{i,W_{3M}}, \tag{4}$$
$$W_{6M} = 126 \text{ trading days}, \qquad \mathrm{Ret}_{i,6M} = \mathrm{Ret}_{i,W_{6M}}. \tag{5}$$

Volatility (1Y)

Let daily returns over the last year be $(R_{i,t-251}, \ldots, R_{i,t})$. Define:

$$\sigma_i = SD\big(R_{i,t-251}, \ldots, R_{i,t}\big), \tag{6}$$
$$\mathrm{Vol}_i = \sigma_i \cdot \sqrt{252}. \tag{7}$$

50-day moving average for trend

$$MA50_i = \frac{1}{50} \sum_{k=1}^{50} P_{i,t-k+1}. \tag{8}$$

Any missing prices or gaps must be filled/treated upstream; this engine assumes a clean daily series.

1.2 Fundamental data (latest, point-in-time)

For each stock i, use only information that was public strictly before date T.

Required.

- Net income (latest annual / TTM).
- Equity (latest).
- Revenue (latest).
- Revenue (1 year ago).
- Operating profit (latest).
- R&D expense (latest, if reported; else marked missing).
- Debt and equity (for Debt/Equity, used in USP).

Derived raw metrics.

Return on Equity

$$ROE_i = \frac{\text{NetIncome}_{i,\text{latest}}}{\text{Equity}_{i,\text{latest}}}. \tag{9}$$

Revenue growth (YoY)

$$\text{RevGrowth}_i = \frac{\text{Revenue}_{i,\text{latest}} - \text{Revenue}_{i,\text{latest-1Y}}}{\text{Revenue}_{i,\text{latest-1Y}}}. \tag{10}$$

Profit margin

$$\text{ProfitMargin}_i = \frac{\text{OperatingProfit}_{i,\text{latest}}}{\text{Revenue}_{i,\text{latest}}}. \tag{11}$$

R&D intensity

$$\text{RnDIntensity}_i = \frac{\text{RnDExpense}_{i,\text{latest}}}{\text{Revenue}_{i,\text{latest}}}. \tag{12}$$

Debt/Equity (for USP)

$$DE_i = \frac{\text{Debt}_{i,\text{latest}}}{\text{Equity}_{i,\text{latest}}}. \tag{13}$$

1.3 Quarterly data

Use:

$$Q_0 = \text{latest reported quarter (available before } T), \tag{14}$$

$$Q_{-4} = \text{same quarter last year.} \tag{15}$$

Inputs.

- Revenue: $\text{Revenue}_{i,Q_0}$, $\text{Revenue}_{i,Q_{-4}}$.
- EPS: EPS_{i,Q_0}, $\text{EPS}_{i,Q_{-4}}$.
- Operating margin:

$$OM_{i,Q_x} = \frac{\text{OperatingProfit}_{i,Q_x}}{\text{Revenue}_{i,Q_x}}. \tag{16}$$

Derived.

Revenue YoY

$$\text{RevYoY}_i = \frac{\text{Revenue}_{i,Q_0} - \text{Revenue}_{i,Q_{-4}}}{\text{Revenue}_{i,Q_{-4}}}. \tag{17}$$

EPS YoY

$$\text{EPSYoY}_i = \frac{\text{EPS}_{i,Q_0} - \text{EPS}_{i,Q_{-4}}}{\text{EPS}_{i,Q_{-4}}}. \tag{18}$$

Margin level

$$\text{MarginLevel}_i = OM_{i,Q_0}. \tag{19}$$

Margin change YoY

$$\text{MarginChangeYoY}_i = OM_{i,Q_0} - OM_{i,Q_{-4}}. \tag{20}$$

1.4 Sector and qualitative data

Sector classification.

- Sector/industry code s (e.g., GICS/ICB), used for peer sets.

USP checklist (redesigned). We use a two-dimensional USP layer:

$$USP2_i,\ USP3_i \in \{0.2, 0.5, 1.0\}. \tag{21}$$

USP2: profitability consistency (3-year history)

Purpose: reward consistency of profit over the last 3 years. Input: last 3 years net income (profit/loss flags by year).

Scoring:

- $USP2_i = 1.0$ if profitable in all 3 years (no loss year).
- $USP2_i = 0.5$ if exactly 1 loss year (2 profit, 1 loss).
- $USP2_i = 0.2$ if 2 or 3 loss years (structurally weak).

USP3: leverage vs sector-specific threshold

Purpose: reward conservative balance sheet vs sector-specific leverage norms.

- Input: DE_i = Debt/Equity.
- Configuration: sector-specific threshold $DE_{\text{threshold},s}$ for each sector s.

Scoring (relative to $DE_{\text{threshold},s}$):

- $USP3_i = 1.0$ (clearly conservative) if

$$DE_i \leq 0.7 \times DE_{\text{threshold},s}. \tag{22}$$

- $USP3_i = 0.5$ (moderate) if

$$0.7 \times DE_{\text{threshold},s} < DE_i \leq 1.3 \times DE_{\text{threshold},s}. \tag{23}$$

- $USP3_i = 0.2$ (stretched) if

$$DE_i > 1.3 \times DE_{\text{threshold},s}. \tag{24}$$

If either USP2 or USP3 is missing, treat that component as 0.5 (neutral).

Aggregate USP score.

$$USPScore_i = \frac{USP2_i + USP3_i}{2}, \quad USPScore_i \in [0, 1]. \tag{25}$$

1.5 Macro inputs (light regimes)

Two binary macro regime flags (provided by a macro service or simple rules):

$$\text{MacroHV}_T \in \{0, 1\} \quad \text{(High-vol / risk-off, e.g., VIX high, index drawdown)}, \tag{26}$$

$$\text{MacroRO}_T \in \{0, 1\} \quad \text{(Risk-on / strong bull, e.g., positive 6M index return, moderate vol)}. \tag{27}$$

Exact thresholds live in configuration; the engine only consumes these binary flags.

2 Normalization and Missing Data Rules

2.1 Peer set selection

For each metric X_i:

- Preferred peer set: all stocks in sector S_s.
- If sector peer set size $N < 20$, use full universe.

Let N be peer set size.

2.2 Percentile ranking

For a "higher-is-better" metric:

- Take raw values X_i.
- Sort ascending; compute rank $\text{Rank}(X_i) \in \{1, \ldots, N\}$, where 1 is worst, N is best.

Define percentile rank:

$$\text{PctRank}(X_i) = \frac{\text{Rank}(X_i) - 1}{N - 1}. \tag{28}$$

Define score:

$$\text{Score}(X_i) = \text{PctRank}(X_i) \in [0, 1]. \tag{29}$$

For "lower-is-better" metrics (e.g., volatility), use:

$$\text{Score}(X_i) = \text{Score}(-X_i). \tag{30}$$

Guarantees.

- All primitive scores Score$(\cdot)$ lie in $[0, 1]$ by construction.
- All composite scores are built only from sums/averages/products with non-negative weights, so they remain in $[0, 1]$ mathematically.
- Implementation may clip to $[0, 1]$ as a safety check, but clipping is not required by design.

2.3 Global missing data rule

If any metric needed for a composite average is missing:

- Replace its score by 0.5 (neutral) before averaging.
- Apply this rule consistently for all stocks and all factors.

3 Factor Scores

We define the following factor scores:

- FundamentalScore (with R&D).
- QuarterlyScore.
- MomentumScore (with trend confirmation).
- RiskScore.
- USPScore.
- BaselineHealth.

3.1 FundamentalScore (includes R&D)

Inputs: ROE_i, RevGrowth_i, ProfitMargin_i, RnDIntensity_i.

$$ROEScore_i = \text{Score}(ROE_i), \tag{31}$$
$$\text{RevGrowthScore}_i = \text{Score}(\text{RevGrowth}_i), \tag{32}$$
$$\text{ProfitMarginScore}_i = \text{Score}(\text{ProfitMargin}_i), \tag{33}$$
$$\text{RnDScore}_i = \text{Score}(\text{RnDIntensity}_i). \tag{34}$$

Core quality fundamental:

$$QualityFundamental_i = \frac{ROEScore_i + \text{RevGrowthScore}_i + \text{ProfitMarginScore}_i}{3}. \tag{35}$$

Combined FundamentalScore:

$$FundamentalScore_i = 0.75 \cdot QualityFundamental_i + 0.25 \cdot \text{RnDScore}_i. \tag{36}$$

If any component score is missing, plug 0.5 before averaging.

3.2 QuarterlyScore

Inputs: RevYoY_i, EPSYoY_i, MarginLevel_i, MarginChangeYoY_i.

$$\text{RevYoYScore}_i = \text{Score}(\text{RevYoY}_i), \tag{37}$$
$$\text{EPSYoYScore}_i = \text{Score}(\text{EPSYoY}_i), \tag{38}$$
$$\text{MarginLevelScore}_i = \text{Score}(\text{MarginLevel}_i), \tag{39}$$
$$\text{MarginChangeScore}_i = \text{Score}(\text{MarginChangeYoY}_i). \tag{40}$$

Margin trend:

$$MarginTrendScore_i = \frac{\text{MarginLevelScore}_i + \text{MarginChangeScore}_i}{2}. \tag{41}$$

QuarterlyScore:

$$QuarterlyScore_i = \frac{\text{RevYoYScore}_i + \text{EPSYoYScore}_i + MarginTrendScore_i}{3}. \tag{42}$$

3.3 MomentumScore

Inputs: $\text{Ret}_{i,1M}$, $\text{Ret}_{i,3M}$, $\text{Ret}_{i,6M}$, $MA50_i$, $P_{i,T}$.

Raw momentum (medium-term emphasis):

$$MomentumRaw_i = 0.2 \cdot \text{Ret}_{i,1M} + 0.4 \cdot \text{Ret}_{i,3M} + 0.4 \cdot \text{Ret}_{i,6M}. \tag{43}$$

Base momentum score:

$$BaseMomentumScore_i = \text{Score}(MomentumRaw_i). \tag{44}$$

Trend filter:

$$TrendFilter_i = \begin{cases} 1, & P_{i,T} > MA50_i, \\ 0, & \text{otherwise.} \end{cases} \tag{45}$$

Final MomentumScore:

$$MomentumScore_i = BaseMomentumScore_i \cdot (0.5 + 0.5 \cdot TrendFilter_i). \tag{46}$$

Interpretation:

- If $P_{i,T} > MA50_i$ (price above 50-day MA) $\Rightarrow$ full momentum score.
- If $P_{i,T} \leq MA50_i$ (price below 50-day MA) $\Rightarrow$ half momentum score (avoid chasing dead bounces).

3.4 RiskScore

Input: Vol_i.

Lower volatility is better, so:

$$RiskScore_i = \text{Score}(-\text{Vol}_i). \tag{47}$$

3.5 USPScore (company strength / moat proxy)

Using profitability and leverage checklist:

- $USP2_i$: profitability consistency (3-year history) scored in $\{0.2, 0.5, 1.0\}$.
- $USP3_i$: conservative leverage vs sector norm scored in $\{0.2, 0.5, 1.0\}$.

If either component is missing, treat it as 0.5. The final USP score is:

$$USPScore_i = \frac{USP2_i + USP3_i}{2}. \tag{48}$$

By construction, $USPScore_i \in [0, 1]$.

3.6 BaselineHealth (absolute checks)

We enforce minimum baselines to avoid pure relative ranking.

Profitability floor

$$BaseProfitFlag_i = \begin{cases} 1, & ROE_i \geq 0, \\ 0, & ROE_i < 0. \end{cases} \tag{49}$$

Growth floor

$$BaseGrowthFlag_i = \begin{cases} 1, & \text{RevGrowth}_i \geq 0, \\ 0, & \text{RevGrowth}_i < 0. \end{cases} \tag{50}$$

Margin floor

$$BaseMarginFlag_i = \begin{cases} 1, & \text{ProfitMargin}_i \geq 0, \\ 0, & \text{ProfitMargin}_i < 0. \end{cases} \tag{51}$$

Aggregate BaselineHealth

$$BaselineHealth_i = \frac{BaseProfitFlag_i + BaseGrowthFlag_i + BaseMarginFlag_i}{3}. \tag{52}$$

By construction, $BaselineHealth_i \in [0, 1]$ and is used as a scalar penalty.

4 Combined Business Quality Layer

4.1 BusinessQualityScore

Combine quantitative fundamentals and USP:

$$BusinessQualityScore_i = 0.8 \cdot FundamentalScore_i + 0.2 \cdot USPScore_i. \tag{53}$$

4.2 Baseline-adjusted Business Quality

$$AdjBusinessQuality_i = BaselineHealth_i \cdot BusinessQualityScore_i. \tag{54}$$

If baselines are violated (negative ROE, negative growth, negative margins), the business quality is proportionally scaled down.

5 Macro Overlay and Weights

Base decision weights (neutral regime):

$$w_F^{(\text{base})} = 0.45, \tag{55}$$

$$w_M^{(\text{base})} = 0.45, \tag{56}$$

$$w_R^{(\text{base})} = 0.10. \tag{57}$$

Macro regime flags: MacroHV_T, MacroRO_T.

5.1 Regime-dependent weights

We define (w_F, w_M, w_R) as follows:

Regime	w_F	w_M	w_R
Neutral (MacroHV = 0, MacroRO = 0)	0.45	0.45	0.10
High-vol / risk-off (MacroHV = 1)	0.45	0.35	0.20
Risk-on bull (MacroRO = 1, MacroHV = 0)	0.40	0.50	0.10
Conflict (MacroHV = 1, MacroRO = 1)	0.45	0.35	0.20

In the conflict case (both flags = 1), we default to the more defensive high-volatility regime. The weights always sum to 1 and are deterministic given the macro flags.

6 Final Score Combination

6.1 FundamentalComposite

We combine adjusted business quality and quarterly trend:

$$FundamentalComposite_i = 0.6 \cdot AdjBusinessQuality_i + 0.4 \cdot QuarterlyScore_i. \tag{58}$$

6.2 DecisionScore

Given regime-dependent weights (w_F, w_M, w_R):

$$DecisionScore_i = w_F \cdot FundamentalComposite_i + w_M \cdot MomentumScore_i + w_R \cdot RiskScore_i. \tag{59}$$

By construction:

- $FundamentalComposite_i, MomentumScore_i, RiskScore_i \in [0, 1]$.
- $w_F + w_M + w_R = 1$ for every regime.

Hence:

$$DecisionScore_i \in [0, 1]. \tag{60}$$

7 Action Mapping and Churn Control

7.1 Action mapping

We map the DecisionScore to a deterministic recommendation:

- If $DecisionScore_i > 0.70$: **BUY**.
- If $DecisionScore_i < 0.35$: **SELL**.
- Otherwise: **HOLD**.

7.2 Churn control

Let the previous state at time T_{prev} be:

$$\left(DecisionScore_i^{(\text{prev})}, Rec_i^{(\text{prev})}\right). \tag{61}$$

Only update the recommendation if:

$$\left|DecisionScore_i - DecisionScore_i^{(\text{prev})}\right| > 0.10 \quad \text{or} \quad (T - T_{\text{prev}}) \geq 14 \text{ days}. \tag{62}$$

Otherwise, keep the previous recommendation $Rec_i^{(\text{prev})}$.

8 Field Definitions and API Contract

This section defines the implementation-level data contract. It does not change the mathematical model in Sections 1–7.

8.1 Per-stock input fields (as of date T)

- Prices:
  - Daily adjusted close $P_{i,t}$ for $t \in [T-251, T]$ (252 trading days).
- Traded value:
  - Daily traded volume $\times$ price (for liquidity checks).
- Fundamentals (latest, point-in-time before T):
  - Net income (latest annual / TTM).
  - Equity (latest).
  - Revenue (latest).
  - Revenue (1 year ago).
  - Operating profit (latest).
  - R&D expense (latest, if reported; else missing).
  - Debt (latest).
- Quarterly data:
  - For Q_0 and Q_{-4}:
    - Revenue: $\text{Revenue}_{i,Q_0}$, $\text{Revenue}_{i,Q_{-4}}$.
    - EPS: EPS_{i,Q_0}, $\text{EPS}_{i,Q_{-4}}$.

    - Operating profit: $\text{OperatingProfit}_{i,Q_0}$, $\text{OperatingProfit}_{i,Q_{-4}}$.

- Sector:
  - GICS/ICB sector code s.
- Macro flags:
  - MacroHV (high-vol / risk-off).
  - MacroRO (risk-on / bull).

8.2 Derived fields (computed from inputs)

Returns.

- Daily simple returns $R_{i,t}$.
- 1M, 3M, 6M cumulative returns: $\text{Ret}_{i,1M}$, $\text{Ret}_{i,3M}$, $\text{Ret}_{i,6M}$.

Volatility and trend.

- $\sigma_i = SD(R_{i,t-251}, \ldots, R_{i,t})$.
- $\text{Vol}_i = \sigma_i \cdot \sqrt{252}$.
- $MA50_i = \frac{1}{50} \sum_{k=1}^{50} P_{i,t-k+1}$.

Fundamental ratios.

- ROE_i (Return on Equity).
- RevGrowth_i (Revenue growth YoY).
- ProfitMargin_i (Operating profit / Revenue).
- RnDIntensity_i (R&D / Revenue).
- DE_i (Debt/Equity).

Quarterly ratios.

- RevYoY_i.
- EPSYoY_i.
- MarginLevel_i.
- MarginChangeYoY_i.

Momentum raw.

- $$MomentumRaw_i = 0.2 \cdot \text{Ret}_{i,1M} + 0.4 \cdot \text{Ret}_{i,3M} + 0.4 \cdot \text{Ret}_{i,6M}. \tag{63}$$

USP components.

- $USP2_i$: profitability score (0.2, 0.5, 1.0).
- $USP3_i$: leverage score (0.2, 0.5, 1.0).
- $USPScore_i = (USP2_i + USP3_i)/2$.

Scores (all in $[0,1]$).

- $ROEScore_i$, RevGrowthScore_i, $\text{ProfitMarginScore}_i$, RnDScore_i.
- $FundamentalScore_i$.
- RevYoYScore_i, EPSYoYScore_i, $\text{MarginLevelScore}_i$, $\text{MarginChangeScore}_i$.
- $MarginTrendScore_i$.
- $QuarterlyScore_i$.
- $BaseMomentumScore_i$, $TrendFilter_i$, $MomentumScore_i$.
- $RiskScore_i$.
- $USPScore_i$.
- $BusinessQualityScore_i$.
- $BaselineHealth_i$.
- $AdjBusinessQuality_i$.
- $FundamentalComposite_i$.
- $DecisionScore_i$.

Regime-dependent weights.

- w_F, w_M, w_R as functions of $(\text{MacroHV}, \text{MacroRO})$ per Section 5.

8.3 Output structure (JSON)

At API level, a typical per-stock response can be structured as:

```
{
  "ticker": "string",
  "date": "YYYY-MM-DD",
  "decision_score": "float [0,1]",
  "recommendation": "string (\"BUY\"|\"HOLD\"|\"SELL\")",
  "factor_breakdown": {
    "fundamental_composite": "float [0,1]",
    "momentum_score": "float [0,1]",
    "risk_score": "float [0,1]",
    "business_quality_score": "float [0,1]",
    "adj_business_quality": "float [0,1]",
    "quarterly_score": "float [0,1]",
    "fundamental_score": "float [0,1]",
    "usp_score": "float [0,1]",
    "baseline_health": "float [0,1]"
  },
  "macro_weights": {
    "w_fundamental": "float",
    "w_momentum": "float",
    "w_risk": "float",
    "macro_hv": "integer (0|1)",
```

```
    "macro_ro": "integer (0|1)"
  },
  "derived_fields": {
    "momentum_raw": "float",
    "trend_filter": "integer (0|1)",
    "returns_1m": "float",
    "returns_3m": "float",
    "returns_6m": "float",
    "volatility": "float",
    "roe": "float",
    "rev_growth": "float",
    "profit_margin": "float",
    "rd_intensity": "float",
    "de_ratio": "float",
    "rev_yoy": "float",
    "eps_yoy": "float",
    "margin_level": "float",
    "margin_change_yoy": "float"
  }
}
```

All scores and ratios should be output with consistent rounding (e.g., 4 decimal places), and missing fields represented as `null`.